Practice workbook · Pixels, resolution and colour depth

Concepts of bitmap representation: what a pixel is, measures of resolution, and why more colour depth makes a bigger file.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-10

1. Question 3 · part b.ii · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.ii only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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1. Question 3 · part b.ii · 2 marks

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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P 7 2 9 3 7 A 0 9 2 4

1. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 1 · part g.i · 1 mark

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Further practice · Attempt part g.i only.

1 Binary digits (bits) are grouped together to represent different types of data.

(a) Hexadecimal notation is sometimes used to represent patterns of binary digits.

Identify **one** reason why programmers use hexadecimal notation.

(1)

- ☒ **A** Easier for a computer to understand.
- ☒ **B** Easier for humans to read.
- ☒ **C** Takes up less computer memory.
- ☒ **D** Quicker than binary to execute.

(b) Convert the bit pattern 0101 1010 to hexadecimal.

(2)

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(c) Identify the number of binary patterns that can be represented by 8 bits.

(1)

- ☒ **A** 128
- ☒ **B** 256
- ☒ **C** 512
- ☒ **D** 1024

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2. Question 1 · part g.i · 1 mark

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Further practice · Attempt part g.i only.

(d) Complete the table by adding these two 8-bit binary integers.

(2)

0	0	1	1	0	0	1	0
0	0	1	1	0	0	1	1

(e) Give the result of performing a logical left shift of 1 place on the binary integer 0100.

(1)

.....

(f) Give the number of bits per character used by standard ASCII.

(1)

.....

(g) **Figure 1** shows a black and white bitmap image.

The pixels in row 5 are represented by the binary pattern 1001 1001

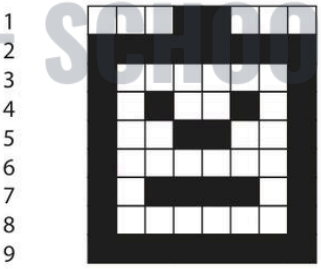


Figure 1

(i) State what is meant by the term **pixel**.

(1)

.....

(ii) Construct an expression to show the number of pixels in the image.

(1)

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2. Question 1 · part g.i · 1 mark · Working and answer

Working and answer

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3. Question 3 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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P 7 2 9 3 7 A 0 8 2 4

3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

4. Question 3 · part b.i · 1 mark

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Further practice · Attempt part b.i only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

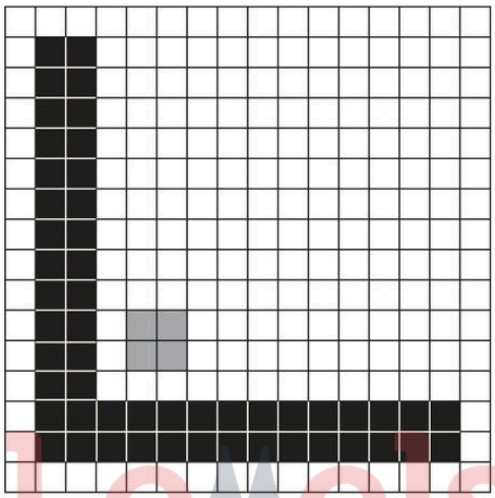


Figure 1

(i) State the smallest component of a bitmap image.

(1)

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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P 7 8 6 3 7 A 0 8 2 0

4. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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